Back-Of-The-Envelope Estimation

#Latency

- Memory is fast but the disk is slow
- Avoid disk seeks if possible
- Simple compression algorithms are fast.
- Compress data before sending it over the internet if possible.

Example:

Assumptions

- 300 million monthly active users.
- 50% of users use daily
- Users post 2 tweets per day on average.
- 10% of tweets contain media
- Data is stored for 5 years

Estimations:

① QPS (Query Per Second)

- DAU = 300m * 50% = 150m
- Tweet QPS = 150m * 2 tweets / 24hour / 3600 second = ~3500
- Peak QPS = 2 * QPS = ~7000

② Storage

- Media storage: 150M * 2 * 10% * 1MB = 30TB per day
- 5 yrs storage: 30TB * 365 * 5 = ~55PB